

NICHOLAS LAM

+65 9151 6192 | nicholaslamzt@gmail.com | <https://www.linkedin.com/in/nicholaslamzt/>
<https://nicholaslamzt.vercel.app>

EDUCATION

Singapore Management University

Singapore

Bachelor of Science (Information Systems)

Aug 2023 - May 2027 (Expected)

Product Development Track

TECHNICAL SKILLS

- Programming Languages: Python, HTML, CSS, Javascript, Typescript, SQL, Java, JCL
- Tools & Frameworks: React, Vue.js, Bootstrap, Next.js, Tailwind CSS, OpenCV, SpringBoot, Expo, Mainframe
- Databases: MySQL, Supabase, Firebase, Firestore, DB2
- Software Development Tools: Figma, Vercel, Netlify, Docker, Postman, Ollama, Azure, AWS, Flask, Anything LLM, Kubernetes, Outsystems
- Design Software: Photoshop, Illustrator, Premiere Pro, Blender

EXPERIENCE

CPF Board

Application Developer Intern

May 2026 – Present

- Designed and developed production-grade services in Java, complete with unit tests, consumed by multiple departments and schemes
- Created batch-level programs using reader-processor-writer architecture with master-slave threading to process account records and update the database at scale
- Participated in Agile/Scrum development cycles, contributing to sprint planning, stand-ups and iterative delivery
- Partnered with cross-functional stakeholders to gather and translate requirements into new features and enhancements for existing systems
- Built and deployed ad-hoc and scheduled batch jobs (JCL) to support critical business processes
- Led SIT and UAT efforts, including test data preparation using SQL and DB2 to ensure release readiness
- Prepared technical documentation for developed features and codebases along with deployment plans to support handover and maintainability
- Refactored legacy codebases while safeguarding functionality through comprehensive regression testing
- Authored Python scripts for data analysis and system concurrency performance

PROJECTS

Dungeons End

Docker Project

Jan 2025 - April 2025

- Text-based dungeon crawler game using Docker containers, adhering to microservice architecture principles. Included inventory management, character stats, and combat mechanics.
- Developed a text-based dungeon crawler using a modular microservice architecture (14+ Python/Flask services) orchestrated via Docker Compose, RabbitMQ, and MySQL.
- Engineered a resilient web gateway and composite service flows enabling login, traversal, combat, inventory, scoring, and session management with robust error handling.
- Implemented asynchronous logging and real-time monitoring, persisting audit trails in MySQL via RabbitMQ while ensuring a responsive retro jQuery/Flask UI with combat logs, health bars, and stat panels.

Passport photo maker

Java Project

Jan 2025 - April 2025

- Built a full-stack web app (Java/Spring Boot, React/Tailwind) that removes photo backgrounds using U2Net (ONNX Runtime) and applies adjustments (brightness/contrast/saturation)
- Engineered a high-quality image pipeline combining deep learning, OpenCV morphological refinement, and GrabCut to achieve crisp alpha mattes and professional compositing across diverse skin tones and clothing.